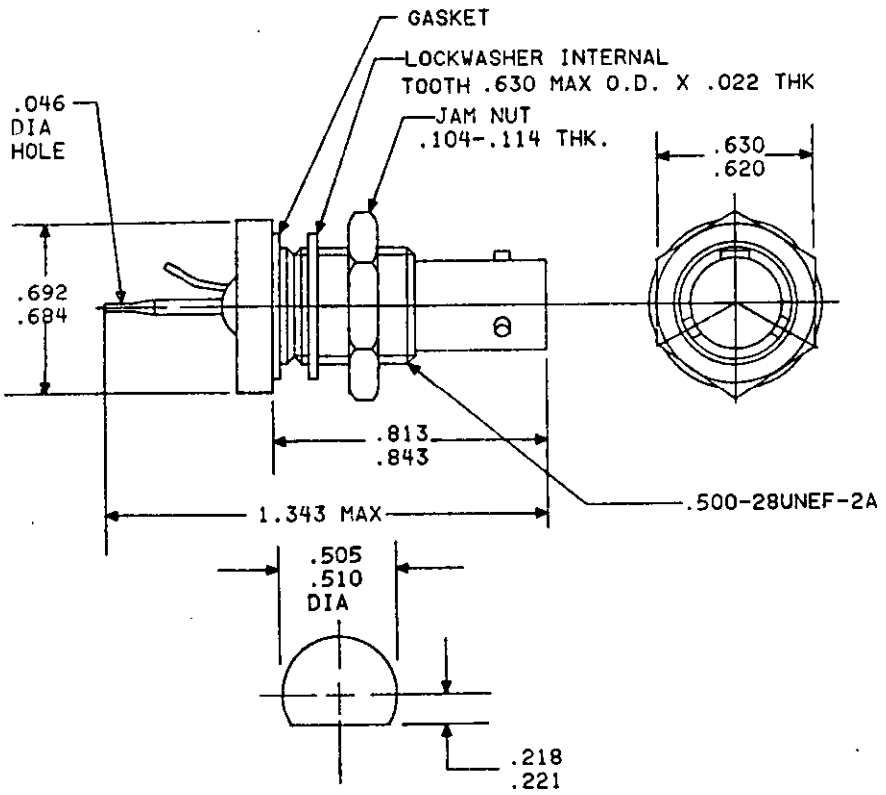
MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY, (SERIES TRB, SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2) HERMETIC AND NON HERMETIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-49142.



MOUNTING HOLE

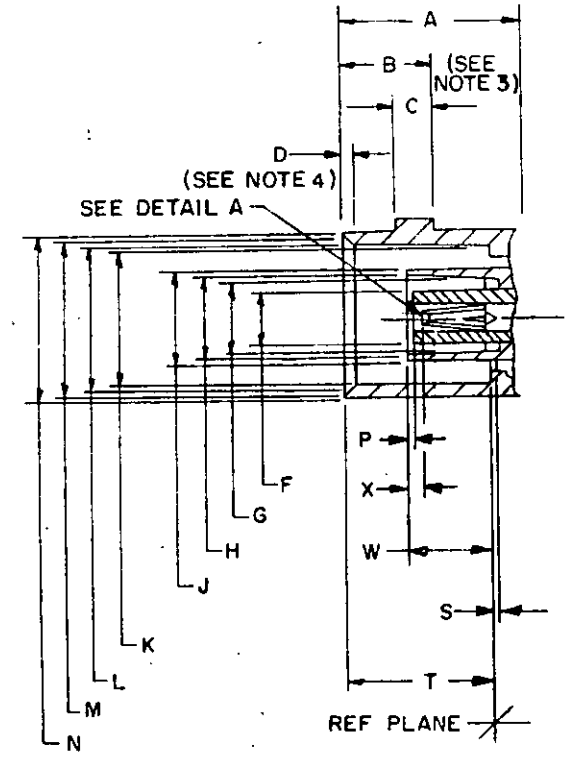
INCHES	MM
.022	.56
.046	1.17
.104	2.64
.114	2.90
.218	5.54
.221	5.61
.500	12.70
.505	12.71
.510	12.95
.620	15.75
.630	16.00
.684	17.37
.692	17.58
.813	20.65
.843	21.41
1.343	34.11

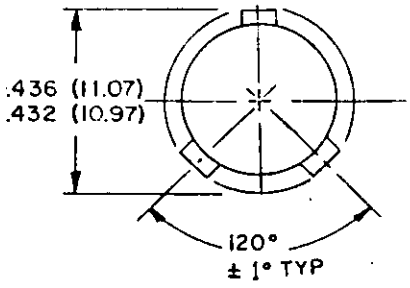
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Wrench flats are to accommodate standard wrench openings per H-28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Panel thickness, .045 (1.14 mm) min, .240 (6.09 mm) max.

FIGURE 1. General configuration.

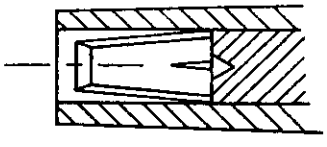
FSC 5935

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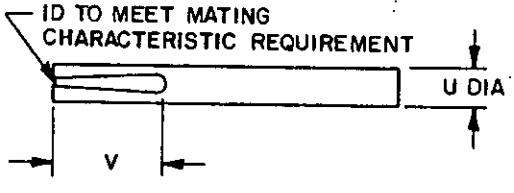




LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.414	---	10.52	---
B	.204	.208	5.18	5.28
C	.075	.081	1.90	2.06
D	.015	.030	.38	.76
F	.117	.122	2.97	3.10
G	.162	.168	4.11	4.26
H	.178	.182	4.52	4.62
J	.195	.199	4.95	5.05
K	.319	.321	8.10	8.15
L	.327	.333	8.31	8.46
M	.346	.360	8.79	9.14
N	.378	.382	9.60	9.70
P	.000	.015	.00	.38
S	.001	---	.03	---
T	.327	.335	8.31	8.51
U	.062	.064	1.57	1.63
V	.213	---	5.41	---
W	.191	.213	4.85	5.41
X	.010	.038	.25	.97



ALTERNATE CLOSED ENTRY INSULATOR
DETAIL A



CONTACT

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. .005 (.13 mm) flat permissible to meet dimension B.
4. Chamfer or radius.
5. Metric equivalents are in parentheses.

FIGURE 2. Mating dimensions for socket contact termination.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Force to engage and disengage:
Longitudinal force - 4 pounds maximum.
Torque - 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See figure 2 for dimensions.

Center contact (socket):
Oversize test pin - 0.040 inch diameter minimum (nonclosed entry contacts only).
Insertion depth - .125 inch minimum.
Number of insertions - 1.

Insertion force test:
Steel test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.
Steel test pin diameter - .039 minimum +.001 inch.

Withdrawal force test:
Steel test pin diameter - .037 maximum -.001 inch.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Permeability: Applicable.

Hermetic seal: See table I.

Leakage: Connector shall be mounted in mounting hole shown on figure 1 with mating end capped. Air pressure - 30 lb_f/in^2.

Insulation resistance: 5000 megohms.

Conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage:. At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Corrosion (salt spray): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

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Conductor resistance: In milliohms maximum.

	Connector dash number	Initial	After environment
Center conductor	0002	2.0	2.5
	0003	4.4	4.9
Intermediate conductor	0002	0.5	0.6
	0003	1.0	1.5
Outer conductor	0002	0.5	0.6
	0003	0.5	0.6

Corona level:
Altitude - 70,000 feet.
Voltage - 200 V rms minimum.

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention: Not applicable.

Coupling mechanism: Not applicable.

Rise time degradation: Not applicable.

Connector durability: 500 cycles minimum, at 12 cycles per minute maximum.

Part number M49142/04- (Dash number from table I.)

Not for Naval Air Systems Command (AS) use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Cross reference of part numbers.

Dash number	Superseded part number
-0002	M49142/04-0001
-0003 (hermetic seal)	

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
Army - CR

(Project 5935-3372)

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC
Air Force - 19

Agent:
DLA-ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49142/4C	Connectors, Recpt, Elec, Triax, RF

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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